# Fundamental Circuit Theory Theorems

*Description & Objectives:*

Three theorems fundamental to the study of circuit theory are explored in this laboratory. The first two are classic engineering models designed to represent lumped linear non-ideal voltage and current sources. These are known as the Thevenin equivalent circuit for the non-ideal voltage source, and Norton equivalent circuit for the non-ideal current source. We will then use these engineering models to consider what optimum load resistance is required for maximum power to flow from a non-ideal source into a resistive load that experimentally confirms the maximum power transfer theorem.

*General Discussion:*

**Thevenin and Norton Engineering Models**

The basic toolbox needed for lumped circuit analysis includes only *ideal* linear components and elements. These include: passive resistors, capacitors, inductors, voltage and current sources. Non-ideal lumped circuits can then be designed with suitably constructed engineering models composed only of these basic ideal linear blocks[1]. Probably the most fundamental of these are the Thevenin and Norton equivalent circuits shown in Fig. 1.

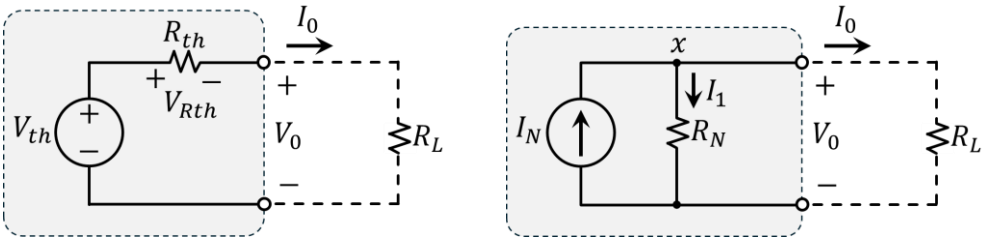



Fig. 1. Fundamental Thevenin (left) and Norton (right) equivalent circuits.

The Thevenin model accounts for the fact that a voltage source's output voltage, $V_0$, varies with supplied current by adding a series resistor, $R_{th}$ to satisfy KVL around the single mesh:$V_{th} = V_{Rth} + V_0$. The Norton model accounts for the fact that a current source's output current, $I_0$, varies with supplied voltage by adding a shunt resistor to satisfy KCL at node $x$: $I_N = I_1 + I_0$. These models are frequently used in circuit theory to simplify or "reduce" otherwise complicated networks into either of these two equivalent circuits. This may involve all or just portions of such networks. Moreover, since it can be proven that the two models are electrically equivalent with respect to whatever is connected as a load, $R_L$, either model can easily be converted into the other by *source transformation*.

**Maximum Power Transfer**

Power flowing from an *ideal* source to a resistive load always transfers all of its generated power into the load itself, i.e. $P_{source} = P_{load}$. The actual power is set by the value of the load resistance, but the generator will always supply all of it. Practical non-ideal sources, however, have internal source resistances that inevitably dissipate some of the generated power that would otherwise flow into the load, i.e, $P_{source} = P_{loss} + P_{load}$. Since the power division between the load and internal loss is determined solely by the source and load resistances, the practical question arises regarding what these need to be to realize maximum power flowing into the load *when we are unable to otherwise change the source resistance*. This is especially true for DC sources, but for AC sources transformers can usually be used to change the source resistance. An audio amplifier, for example, driving a 4Ω or 8Ω speaker must consider this question. So, given a fixed load resistance, what must the source resistance be for maximum transfer of power? The answer is given by a famous fundamental theorem of electric circuits. The *maximum power transfer theorem* tells us how to *match* source and load resistances. Later you will see this theorem holds for AC circuits as well, where resistances are generalized as impedances.

[1] Any circuit containing only linear devices and elements is itself also linear. This is a useful feature exploited by the superposition theorem.

The maximum power transfer theorem states that the largest amount of power that can be coupled from a non-ideal source into a load occurs when the load resistance is equal to the source resistance (the derivation is given in section [4.8] of our textbook). To help you see this, consider the following non-ideal voltage source with a fixed and unchangeable source resistance, $R_{th}$, connected to a resistive load, $R_L$:

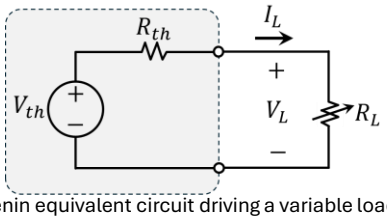



Fig. 2. Thevenin equivalent circuit driving a variable load resistance, $R_L$.

Now find $R_L$ such that maximum power is transferred to the load from the source. The power in the load is given by

$$P_L = I_L \cdot V_L.$$

Noting that

$$V_L = V_{th}\frac{R_L}{R_{th}+R_L}$$

by the voltage divider theorem, and that

$$I_L = \frac{V_{th}}{R_{th}+R_L}$$

by Ohm's Law, we can then write the power in the load as;

$$P_L = \left(\frac{V_{th}}{R_{th}+R_L}\right)V_{th}\frac{R_L}{R_{th}+R_L} = \frac{V_{th}^2 R_L}{(R_{th}+R_L)^2}.$$

Differentiating this equation with respect to $R_L$and setting this derivative equal to zero gives the function's maximum, or the point at which maximum power is transferred.

$$\therefore \frac{dP_L}{dR_L} = 0 = V_{th}^2\left(\frac{(R_{th}+R_L)^2 - 2R_L(R_{th}+R_L)}{(R_{th}+R_L)^4}\right) = V_{th}^2\left(\frac{R_{th}-R_L}{(R_{th}+R_L)^3}\right) = 0.$$

Solving yields

$$R_{th} - R_L = 0, \qquad \text{or;} \quad \boxed{R_L = R_{th}}$$

To gain some visual intuition behind this derivation, if we instead just plot $P_L$ as a function of $R_L$ we can clearly see that when the load resistance equals the source resistance maximum power flows from the source into the load:

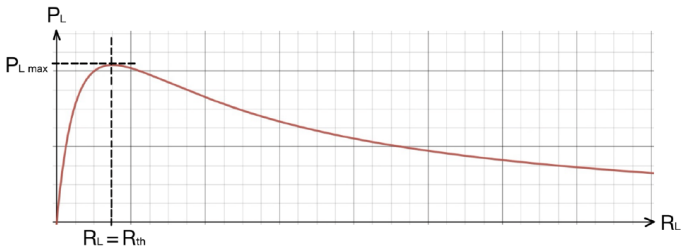



Fig. 3. Plot of power in the load, $P_L$, versus the load resistance, $R_L$.

Thus;

$$P_{L\,max} = \frac{V_{th}^2 R_{th}}{(R_{th}+R_{th})^2} \rightarrow \boxed{P_{L\,max} = \frac{V_{th}^2}{4R_{th}}}$$

We can understand the significance of the somewhat strange cofactor ¼ by appreciating that conceptually $\frac{V_{th}^2}{R_{th}}$ is **not** the total power available, but rather the power dissipated in the internal Thevenin resistance if the load resistance were to be short circuited, i.e. $R_L = 0\Omega$. When the load and source resistances are equal ($R_L = R_{th}$), only half of this power is now available, and this reduced power is itself equally divided between $R_L$ and $R_{th}$.

An additional note on perspective here: There are many cases where what we are interested in is *maximum voltage* across the load, or *maximum current* flowing through it, but not necessarily maximum power delivered to it. As already noted, and evident from Fig.3, maximum power occurs when $\frac{R_L}{R_{th}} = 1$; this is when the volt-amp *product* is largest, but not necessarily the voltage or current alone. It should further be evident that maximum voltage across the load occurs when this ratio is much greater than 1, and maximum current occurs when it is much less than 1. Examples are voltage amplifiers and current amplifiers. Power amplifiers require matching, since they are interested in maximum power transfer, while voltage and current amplifiers do not.

*Experimental Work:*

**Part 1: Thevenin and Norton Equivalent Circuits**

Reconstruct the same 2-port T-network that you used in Lab-1 and devise an experiment to determine the Thevenin and Norton equivalent circuits when driven at the input port by an ideal 10 V DC power source. Perspective is important here. Treat the power source and T-network as if they were combined together inside a mystery "black box" having only one port, an output port. Thus you can experimentally work only with the two nodes at the output port, $a$ and $b$, and whatever is connected to it (e.g. a load $R_L$, an open circuit, a short circuit, digital multimeter, etc.). Fig. 4 shows the general idea.

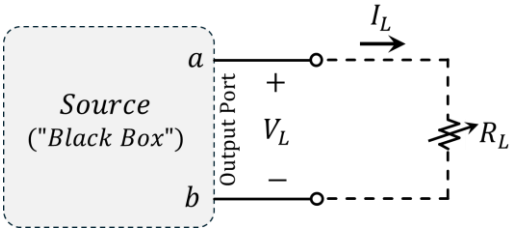



Fig. 4. "Black Box" 1-port source network (Note: a single port comprises two connections).

Since the source transformation theorem relies on the fundamental equivalence of the two models, you should be able to deduce either model solely from your data. After you have experimentally determined both circuits, add a variable load, or use data already taken (with different discrete values for $R_L$), to analytically show that both engineering models can indeed be used to accurately predict any load current and voltage, $I_L$ and $V_L$ within the range permitted by the source.

**CHECK-OFF [1/3] - Verify correct construction and initial data acquisition with the TA.**

**Part 2: Non-Ideal Power Source**

Now let us consider the practical question of just how ideal the lab supply connected above to your T-network really is. Assess this by proceeding to experimentally determine what the power supply's **Thevenin source resistance, $R_{th}$,** is for this experiment that must include the wiring between the supply and your T-network. Be sure to determine this by taking all measurements *at the point the power supply is connected to your breadboard*. Just how ideal is it, and generally, does it need to be taken into account? Be sure to draw an equivalent engineering schematic in your notes clearly showing how you understand this problem.

**CHECK-OFF [2/3] – Verify recorded data and its relevance with the TA.**
**(Be clear on how you will obtain $R_{th}$ experimentally).**

**Part 3: Maximum Power Transfer**

Using either the Thevenin or Norton models you experimentally obtained in Part 1, prove the maximum power transfer theorem experimentally.

**CHECK-OFF [3/3] – Demonstrate relevant collected data to the TA.**

*Report and Submission*

Your Lab-2 Lab Report is due on Nov 13th at 5pm. Please upload it as a .pdf file to Gradescope. Also ensure that the first page of your report is a filled out copy of the **Lab_Report_Cover_Sheet**, located at Canvas / Files / Lab Resources.